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CO4 – AT3
MLA 0201 – Fundamentals of Machine Learning

Annexure A

Comparative Analysis

Question 1 (20 Marks)

A healthcare organization is developing an intelligent disease diagnosis system using patient symptoms and medical history.

Compare and analyze the suitability of **Bayesian Networks** and **Markov Random Fields (MRFs)** for modeling the relationships among diseases and symptoms. Compare both models with respect to graph structure, conditional independence assumptions, inference process, uncertainty handling, computational complexity, and real-world applicability. Justify which model is more appropriate for this healthcare application.

Question 2 (20 Marks)

An autonomous vehicle company is designing a system to recognize driving activities from continuous sensor data collected over time.

Compare and evaluate the performance of **Hidden Markov Models (HMMs)** and **Conditional Random Fields (CRFs)** for sequence labeling and activity recognition. Compare the models based on state representation, observation modeling, learning approach, prediction accuracy, handling of contextual features, computational efficiency, and scalability. Recommend the most suitable model for the given application with proper justification.

Question 3 (20 Marks)

A financial analytics company is building an AI-based risk prediction system using customer transaction data and historical financial records.

Compare Directed Graphical Models (Bayesian Networks) and **Undirected Graphical Models (Markov Random Fields)** for representing probabilistic dependencies in the system. Evaluate both approaches in terms of graph representation, dependency modeling, inference techniques, learning algorithms, generalization capability, computational requirements, and suitability for financial risk prediction. Conclude which graphical model is more effective for this industrial application and justify your recommendation.

1 — Bayesian Networks vs Markov Random Fields in Healthcare

Introduction

An intelligent disease diagnosis system must represent uncertain relationships between **patient symptoms, medical history, risk factors, and diseases**. Two important graphical models for this purpose are **Bayesian Networks (BNs)** and **Markov Random Fields (MRFs)**. Both can represent probabilistic relationships, but they differ in graph structure, independence assumptions, inference, and computational requirements.

Comparison of Bayesian Networks and MRFs

Parameter	Bayesian Network (BN)	Markov Random Field (MRF)
Graph structure	Directed Acyclic Graph (DAG)	Undirected graph
Relationship representation	Arrows show direction of dependency	Edges show mutual dependency
Conditional independence	Based on d-separation	Based on graph separation
Probability model	Factorizes using conditional probabilities	Factorizes using potential functions
Inference	Bayes' rule, variable elimination, belief propagation	Belief propagation, Gibbs sampling, MCMC, optimization
Uncertainty handling	Very strong; directly represents conditional probabilities	Strong; represents relationships using potentials
Observation of causes	Naturally represents cause-effect relationships	Does not naturally represent causal direction
Computational complexity	Can become expensive with large and highly connected networks	Can also be expensive because of normalization/partition function
Interpretability	High when causal relationships are known	Moderate; relationships are symmetric
Healthcare suitability	Highly suitable for diagnosis and medical decision support	Useful for spatial or strongly interconnected medical data

1. Graph Structure

A Bayesian Network uses a **directed acyclic graph**. For example:

Smoking → Lung Disease → Cough

This structure can represent how one factor influences another.

An MRF uses an **undirected graph**, where relationships have no direction:

Smoking — Lung Disease — Cough

Therefore, Bayesian Networks are more suitable when the healthcare system needs to represent **cause-and-effect relationships**.

2. Conditional Independence Assumptions

Bayesian Networks use **d-separation** to determine conditional independence. For example, once a disease is known, some symptoms may become conditionally independent of certain risk factors.

MRFs use **graph separation**. A variable is conditionally independent of another variable when the graph separates them.

For medical diagnosis, the explicit conditional dependencies of Bayesian Networks are generally easier for doctors and system designers to understand.

3. Inference Process

In Bayesian Networks, inference can be performed using methods such as:

- Variable elimination
- Belief propagation
- Bayesian inference
- Sampling methods

For example, if a patient has fever and cough, the system can calculate:

P(Disease | Fever, Cough, Medical History)

MRFs use techniques such as belief propagation, Gibbs sampling, and Markov Chain Monte Carlo. Inference can become difficult when the graph contains many interconnected variables.

4. Handling Uncertainty

Both models are capable of handling uncertainty.

Bayesian Networks assign probabilities to conditional relationships, making them particularly useful when medical knowledge is available in terms such as:

- Probability of disease given symptoms
- Probability of symptoms given disease
- Risk based on medical history

MRFs represent uncertainty through **potential functions**, which measure how compatible neighboring variables are.

5. Computational Complexity

Both models can have high computational complexity for large graphs. Exact inference in graphical models is generally difficult and can become exponential with respect to the graph's connectivity/treewidth.

MRFs additionally require calculation of a **partition function** for normalization, which can be computationally expensive.

Bayesian Networks can also become expensive when there are many dependencies, but their directed structure often makes medical reasoning easier to organize.

6. Real-World Applicability

Bayesian Networks are widely appropriate for:

- Disease diagnosis
- Medical decision support
- Risk prediction
- Patient monitoring
- Treatment planning

MRFs are more useful when relationships are naturally symmetric or spatial, such as:

- Medical image segmentation
- Tumor-region detection
- Image-based disease analysis
- Spatial relationships between biological variables

Critical Evaluation

The main advantage of a Bayesian Network is its **causal and interpretable structure**. Medical experts can construct relationships between diseases, symptoms, and risk factors using domain knowledge. However, building an accurate network requires reliable probability estimates.

MRFs can model complex interactions between variables without requiring a direction for every relationship. However, their inference and parameter estimation can become computationally expensive, especially for highly connected networks.

Conclusion

For the given healthcare disease diagnosis application, **Bayesian Networks are more appropriate.**

2 — HMMs vs CRFs for Autonomous Vehicle Activity Recognition

Introduction

An autonomous vehicle receives continuous sensor data from sources such as **camera systems, GPS, accelerometers, LiDAR, radar, and vehicle sensors**. The system must identify activities such as driving, braking, turning, stopping, accelerating, or changing lanes over time.

Two important sequence models are **Hidden Markov Models (HMMs)** and **Conditional Random Fields (CRFs)**.

Comparison of HMMs and CRFs

Parameter	HMM	CRF
State representation	Hidden states represent activities	Labels represent activities
Observation modeling	Explicitly models observations given states	Does not require an independent observation model
Learning approach	Generative learning	Discriminative learning
Prediction	Viterbi algorithm	MAP decoding / Viterbi-style dynamic programming
Contextual features	Limited by independence assumptions	Can use many overlapping features
Accuracy	Good for simple sequential patterns	Usually better for complex feature-rich sequences
Computational efficiency	Efficient for small/simple models	More computationally expensive during training
Scalability	Good for simple systems	Better for complex feature-based systems but training can be costly
Main limitation	Strong independence assumptions	Feature engineering and training complexity
Application suitability	Simple activity recognition	Complex sensor-based activity recognition

1. State Representation

In an HMM, the actual activity is a **hidden state**, while sensor readings are observations.

For example:

Driving → Turning → Braking → Stopping

The vehicle's sensors produce observations, while the HMM estimates the hidden driving activity.

A CRF directly models the conditional probability of the complete label sequence given the observed sensor data.

2. Observation Modeling

HMMs use an explicit observation model:

P(Observation | State)

This means that the model must make assumptions about how sensor observations are generated from each activity.

CRFs directly model:

P(Labels | Observations)

Therefore, CRFs do not need to model how the sensor data was generated. This makes them more flexible for complex sensor information.

3. Learning Approach

HMMs are **generative models**. They model both the transition between hidden states and the probability of observations.

CRFs are **discriminative models**. They focus directly on predicting the correct activity labels from observed sensor features.

For autonomous vehicles, this is useful because the main objective is not to generate sensor data but to **accurately classify the vehicle's activity**.

4. Prediction Accuracy

HMMs work well when activities have simple sequential patterns. For example:

Driving → Braking → Stopping

However, real autonomous vehicles have many overlapping features. A braking event may depend on speed, acceleration, distance, road conditions, GPS position, and surrounding vehicles.

CRFs can combine these different features and therefore can provide better prediction accuracy in complex environments.

5. Contextual Features

This is one of the biggest differences.

HMMs generally rely on assumptions such as:

- Current state depends mainly on previous state.
- Current observation depends mainly on current state.

CRFs can use a much wider range of contextual features, such as:

- Current speed
- Previous speed
- Acceleration
- Steering angle
- GPS position
- Distance from other vehicles
- Road condition
- Previous and future observations

Thus, CRFs are more flexible for autonomous vehicle activity recognition.

6. Computational Efficiency

HMMs are computationally efficient because algorithms such as **Forward-Backward** and **Viterbi** can perform inference efficiently.

CRFs also support efficient inference for linear-chain structures, but training can be more expensive because parameters are learned discriminatively and feature combinations may be large.

Therefore:

HMM → simpler and faster

CRF → more flexible but potentially more expensive

7. Scalability

HMMs scale well for simple state spaces, but their performance can decrease when many contextual features are required.

CRFs are better suited for feature-rich sequence labeling, although the number of features and training data can increase computational requirements.

Critical Evaluation

HMMs are attractive because they are simple, interpretable, and computationally efficient. They are suitable when the activity sequence follows clear transition patterns.

However, autonomous driving is a complex environment. Sensor data is affected by road conditions, traffic, speed, weather, and surrounding objects. HMM independence assumptions may therefore be too restrictive.

CRFs can combine multiple contextual features without requiring the same strong independence assumptions. Their major disadvantages are higher training complexity and the need for suitable feature design.

Conclusion

For the given **autonomous vehicle activity recognition system**, **CRFs are more suitable**.

The reason is that autonomous vehicles generate **multiple, noisy, and highly contextual sensor features**. CRFs can use these features together and directly optimize activity-label prediction.

3 — Bayesian Networks vs MRFs for Financial Risk Prediction

Introduction

A financial analytics company may need to predict risks such as **loan default, fraud, credit risk, financial loss, and customer risk** using transaction data and historical financial records. Graphical models can represent dependencies between financial variables and support probabilistic risk prediction.

The two important approaches are **Directed Graphical Models (Bayesian Networks)** and **Undirected Graphical Models (MRFs)**.

Comparison of Bayesian Networks and MRFs

Parameter	Bayesian Network	Markov Random Field
Graph representation	Directed Acyclic Graph	Undirected graph
Dependency direction	Clearly represents direction	Represents symmetric relationships
Probability factorization	Conditional probability distributions	Potential functions
Dependency modeling	Excellent for causal relationships	Good for mutual interactions
Inference	Variable elimination, belief propagation, sampling	Belief propagation, MCMC, Gibbs sampling
Learning	Parameter and structure learning	Parameter learning and feature/potential estimation
Interpretability	High	Moderate
Generalization	Good when causal structure is correct	Good for complex interactions
Computational requirements	Can be high for complex networks	Often high due to normalization and dense dependencies
Financial suitability	Strong for risk and causal analysis	Useful for interconnected transaction patterns

1. Graph Representation

A Bayesian Network uses directed edges.

For example:

Income → Credit Score → Loan Default

This clearly shows the direction of dependency.

An MRF uses undirected edges:

Income — Credit Score — Loan Default

It shows that the variables are related but does not specify which variable causes another.

For financial risk prediction, directed relationships are often useful because analysts want to understand **why a customer is considered risky**.

2. Dependency Modeling

Bayesian Networks can represent relationships such as:

Low Income → Low Repayment Ability → Default Risk

They can also include previous defaults, debt, transaction behavior, and credit history.

MRFs are useful when many variables influence each other simultaneously. For example, transaction amount, transaction location, device information, and account activity may have strong mutual relationships.

However, MRFs do not naturally express causal direction.

3. Inference Techniques

Bayesian Networks use:

- Variable elimination
- Belief propagation
- Sampling
- Bayesian inference

They can calculate probabilities such as:

$P(\text{Default} \mid \text{Income, Credit Score, Previous Default, Transactions})$

MRFs use:

- Gibbs sampling
- Markov Chain Monte Carlo
- Belief propagation
- Optimization-based methods

These methods can become computationally expensive when the graph is large or densely connected.

4. Learning Algorithms

Bayesian Networks use two major types of learning:

Parameter learning: estimating probabilities from historical financial data.

Structure learning: discovering relationships between variables from data.

MRFs learn parameters for their potential functions. In complex applications, feature weights need to be estimated from data.

Bayesian Networks have an additional advantage when domain experts already understand important financial relationships because expert knowledge can be included in the network structure.

5. Generalization Capability

A Bayesian Network can generalize well when the underlying dependency structure is correctly designed. It can also handle missing information and uncertain evidence.

MRFs can generalize well when variables have strong local interactions and the system needs to model complex dependencies.

For financial risk prediction, Bayesian Networks generally provide better **explainability**, which is important because financial institutions often need to explain why a customer was classified as high risk.

6. Computational Requirements

Both models can become computationally expensive for large datasets.

Bayesian Networks may require significant computation when the network has many interconnected variables.

MRFs can be especially expensive because probability normalization requires computing a **partition function**, which may be difficult for large graphs.

Approximate inference methods can reduce the computational cost in both models.

7. Suitability for Financial Risk Prediction

Bayesian Networks are highly suitable for:

- Credit risk prediction
- Loan default prediction
- Fraud risk analysis
- Customer risk assessment
- Financial decision support

MRFs are useful when the main problem involves **complex interactions between interconnected variables**, such as transaction networks and relationships between multiple entities.

Critical Evaluation

The major strength of Bayesian Networks is their **interpretability and causal representation**. Financial analysts can understand how factors such as income, debt, credit history, and previous defaults contribute to risk.

Their weakness is that creating an incorrect network structure can lead to poor predictions. Large networks can also make inference expensive.

MRFs are powerful for representing complex relationships without forcing a direction between variables. This can be useful for transaction networks where relationships are naturally interconnected. However, they are less intuitive for causal financial reasoning and may require more computational resources.

Conclusion

For the given **financial risk prediction application, Bayesian Networks are generally more effective**.

The system needs to combine historical financial records and customer transaction information to estimate the probability of future risk. Bayesian Networks provide:

- Clear dependency representation
- Probabilistic risk estimation
- Better interpretability
- Ability to incorporate expert financial knowledge
- Handling of missing and uncertain data
- Support for causal reasoning